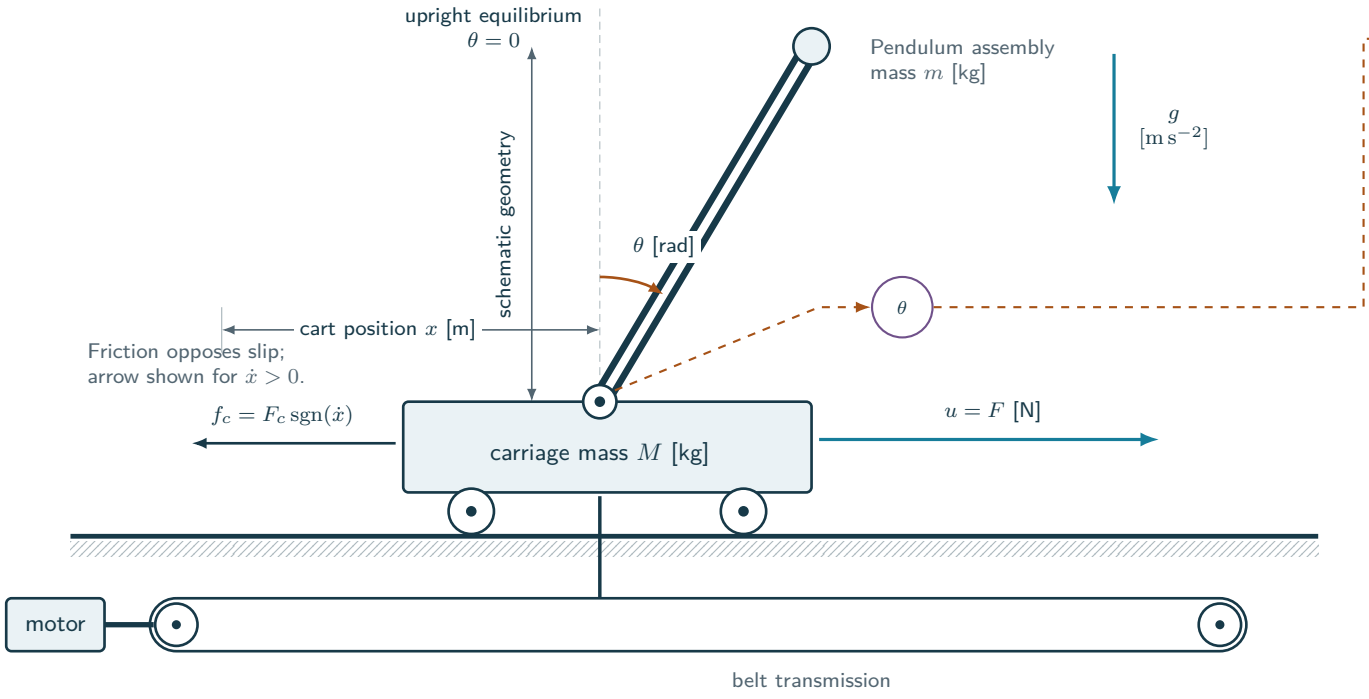
Cart and inverted pendulum

Paper section 3.3: *Cart with inverted pendulum*

Belt-driven carriage and freely rotating pendulum



Control problem

Swing up from the hanging position, balance near the unstable upright state, then position the cart.

Published measurement

Angular position $y = \theta$.

Simulation state order

$[\theta,\ \dot{\theta},\ x,\ \dot{x}]^{\mathsf{T}}$

Units: rad, rad/s, m, m/s.

Nonlinear couplings

Gravity $\sin\theta$, inertial terms $\dot{\theta}^2$, trigonometric coupling and Coulomb friction.

Angular joint loss: $d_{Mf}\dot{\theta}$ [N m].

Input and actuator limit

$u = F$ [N], $|F| \le F_{\max}$.

The source uses a motor and belt; Eq. (32) takes the resulting force as its input.

Length convention

l [m] is the characteristic length used in Eq. (32). The drawing does not infer a separate rod length, inertia or centre of mass from l.

Reading the drawing

Solid arrows: forces / motion.

Dashed arrows: measurement signals.

Geometry is not to scale.

Model scope: Eq. (32) with explicit force saturation and slip friction. No rail-impact or static-sticking model is implied. The educational learning study may declare different friction settings in its case sheet.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.